

HTU2X Serial Number Reading

OVERVIEW

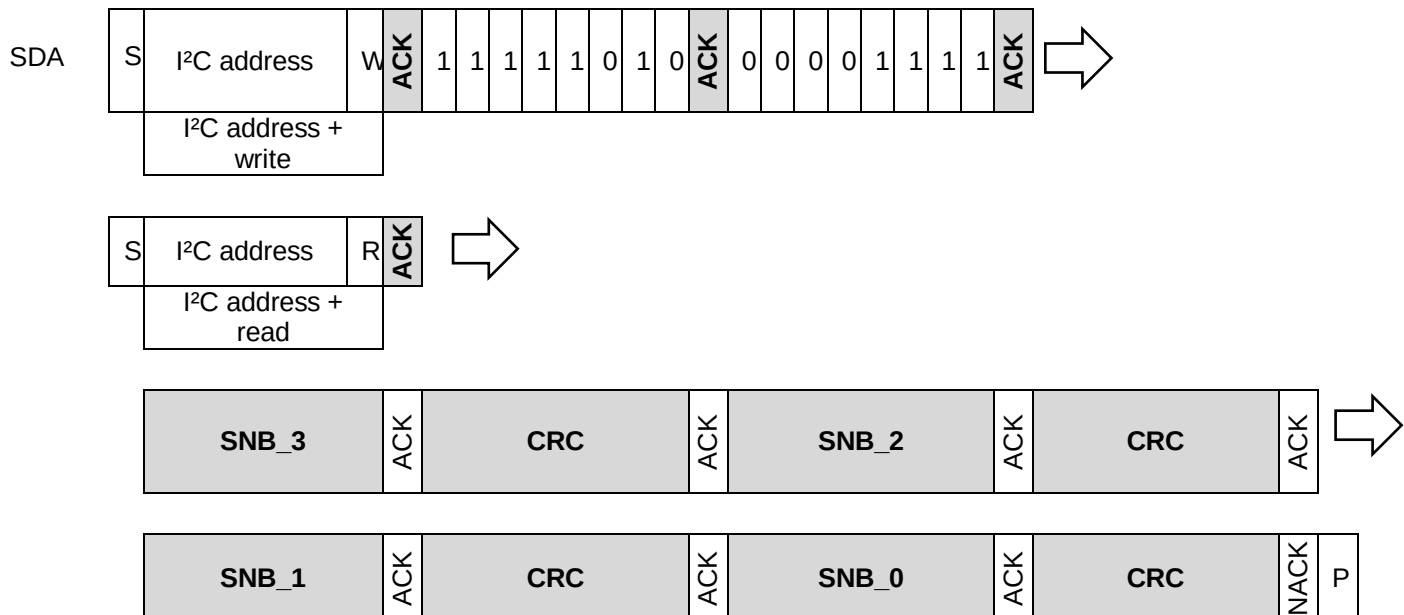
The HTU2X provides a device specific serial number which can be read-out via the serial interface (I²C) and allows for an unambiguous identification of each individual device. This application note describes the procedure for read-out of the serial number.

COMMUNICATION SEQUENCE

The communication sequences for reading out the serial number comply with the general I²C serial interface protocol. For detailed information on I²C, please refer to the Datasheet HTU2XD (HPC199 document).

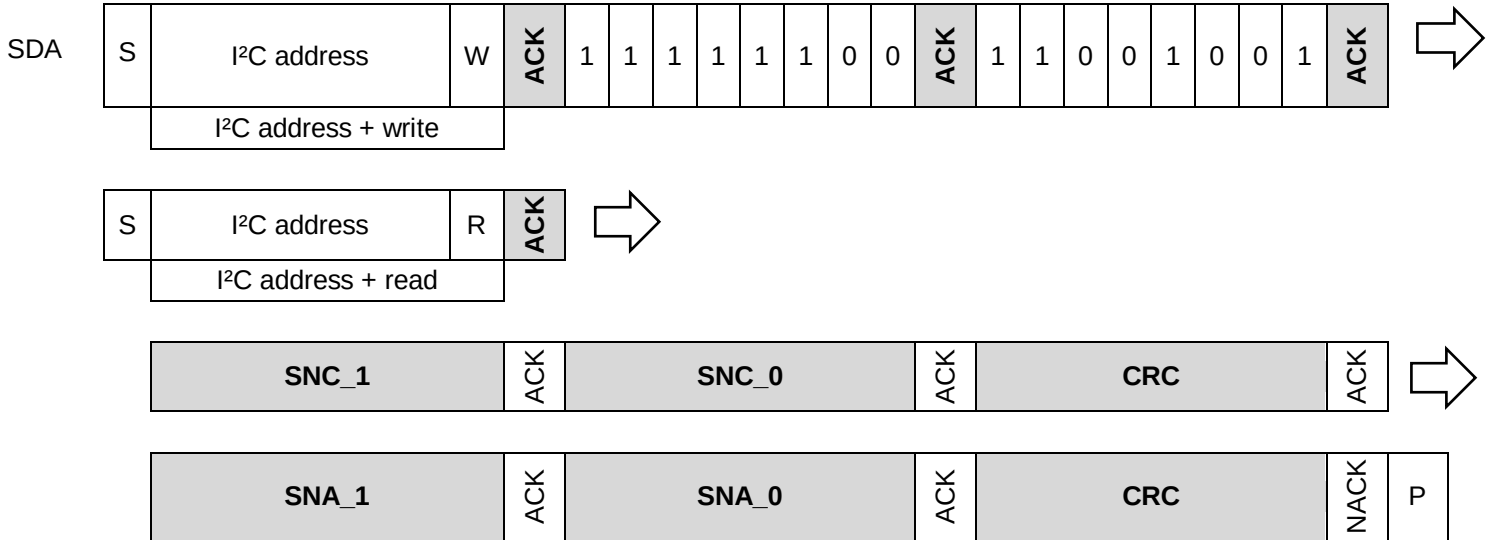
The serial number of the HTU2X is partitioned to two different locations on the on-chip memory. Therefore, two memory access sequences are required to retrieve the complete serial number information. Figure below shows the corresponding communication sequence.

First memory access:



HTU2X Serial Number Reading

Second memory access:



S is for I²C start condition

P is for I²C stop condition

Data in bold are data from HTU2X sensor to master

COMPOSITION OF SERIAL NUMBER

After reading out the data as described above, the serial number is obtained by arranging the data bytes according to figure below.

64 bits							
16 bits		32 bits				16 bits	
SNA_1	SNA_0	SNB_3	SNB_2	SNB_1	SNB_0	SNC_1	SNC_0
0x48*	0x54*	0x00*	0xXX	0xXX	0xXX	0x32*	0xXX

* are fixed values

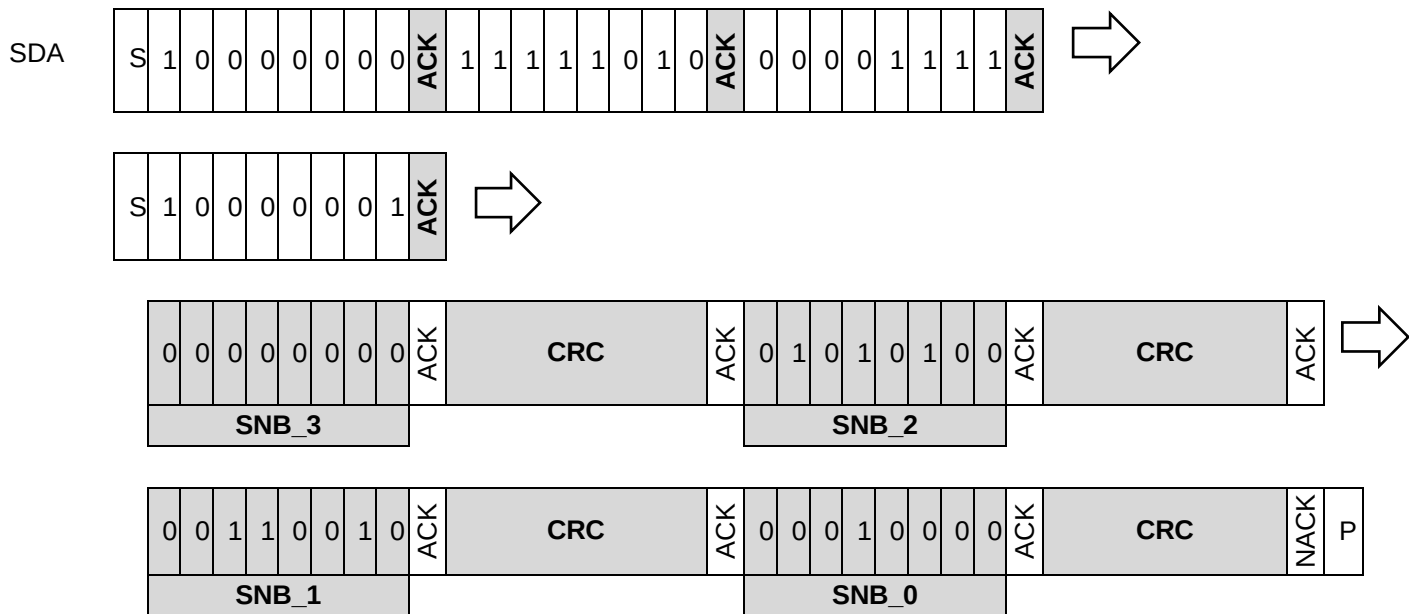
HTU2X Serial Number Reading

EXAMPLE

For illustration of the read-out procedure, a numeric example is presented below. The I²C address of the HTU2X is assumed as "0100'0000" (hexadecimal: 0x40).

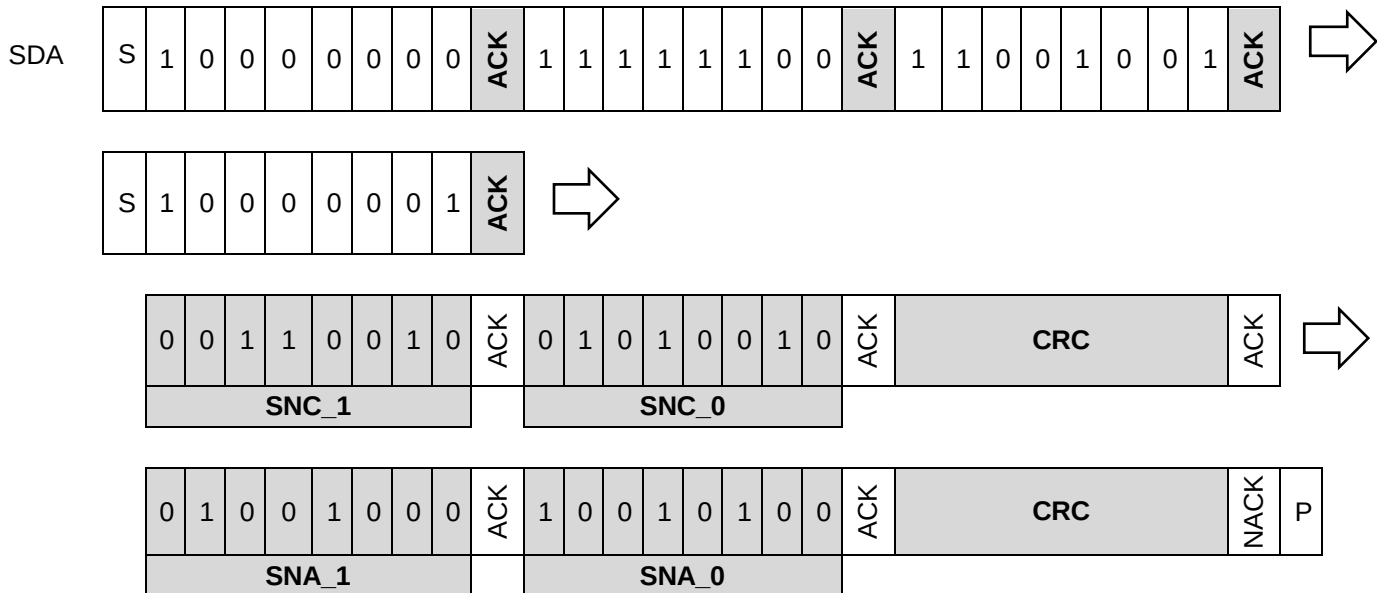
The serial number of the numeric example reads (hexadecimal) is 0x4854'0054'3210'3212.

First memory access:



HTU2X Serial Number Reading

Second memory access:



Customer Service contact details

Measurement Specialties, Inc - MEAS France
 Impasse Jeanne Benozzi
 CS 83 163
 31027 Toulouse Cedex 3
 FRANCE
 Tel:+33 (0)5 820.822.02
 Fax:+33 (0)5.820.821.51
 Sales: humidity.sales@meas-spec.com

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0	Document Creation	D. LE GALL	October 12

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